ENGINEERING RÉSUMÉ

William Anthony

Junior Edge AI Engineer | Microelectronics Research | Digital Design & Verification

Bandung, Indonesia | Institut Teknologi Bandung | Expected graduation: July 2027

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PROFESSIONAL SUMMARY

Electrical and Electronics Engineering student at ITB with current experience as a Junior Edge AI Engineer and a Research Assistant at the Microelectronics Center, plus a completed Product Consumables internship at Alcon. Develops Python-based sensing and AI prototypes, synthesizable SystemVerilog hardware, and production-data automation. Ranked first of 118 students with a GPA of 3.70 / 4.00; named ITB's 2026 Most Outstanding Electrical and Electronics Engineering Student. Combines implementation, verification, measurement, and technical communication across hardware and software.

EDUCATION

Institut Teknologi Bandung Jul 2023 - Jul 2027 (expected)

Bachelor of Engineering in Electrical and Electronics Engineering | GPA: 3.70 / 4.00 | Academic rank: 1 of 118

Relevant coursework: VLSI System Design, Digital IC Analysis and Design, Computer Architecture, Digital Signal Processing, Analog Mixed-Signal Electronic Design, Electronics, and Programming.

PROFESSIONAL EXPERIENCE

Junior Edge AI Engineer, Computer Vision & Audio Sep 2026 - Present

Institut Teknologi Bandung | Part-time | Bandung, Indonesia / Hybrid

Build Python and embedded proof-of-concept systems for real-world audio and visual sensing, from acquisition and preprocessing to model deployment, evaluation, and result analysis.

- Develop microphone-based sensing pipelines with INMP441, I2S, ESP32-S3, USB CDC, and Raspberry Pi.
- Integrate YAMNet and TensorFlow models for environmental sound classification and evaluate model behavior using recorded WAV datasets and synthetic signals.
- Design modular, testable components for acquisition, preprocessing, inference, and analysis; validate behavior across the end-to-end sensing pipeline.
- Evaluate deployment latency, memory use, model caching, and real-time performance on embedded platforms.

Research Assistant - Digital Hardware & Signal Processing Sep 2026 - Present

Microelectronics Center of Institut Teknologi Bandung | Full-time | Bandung, Indonesia / On-site

Engineer and verify synthesizable SystemVerilog hardware for embedded signal-processing applications, translating algorithm requirements into implementation-ready RTL.

- Develop RTL microarchitecture, datapaths, and controllers for synchronous digital signal-processing systems.
- Apply pipelining and resource optimization to balance implementation cost and throughput requirements.
- Build self-checking verification flows and analyze functional behavior and performance before FPGA validation.
- Prepare synthesizable designs for practical embedded-system integration and FPGA implementation workflows.

Product Consumables Intern Jul 2026 - Aug 2026

Alcon | Internship | Batam, Indonesia

Delivered production-data automation during a manufacturing internship, consolidating more than 40 Excel sheets into a standardized operational dataset.

- Consolidated 40+ production Excel sheets into a standardized dataset using Power Apps, Power Automate, and Excel.
- Developed dashboard data flows against user requirements to make production information easier to review and maintain.
- Implemented automated notification flows for production operations and communicated the completed workflow in the final internship presentation.

RESEARCH & TEACHING EXPERIENCE

Electronics II Laboratory Coordinator
Sep 2026 - Present

Institut Teknologi Bandung | Part-time | Bandung, Indonesia / On-site

- Coordinate Electronics II laboratory delivery for 111 students and train and supervise 12 assistants, approximately one assistant per nine students.
- Translate learning outcomes into requirements, circuit constraints, instrumentation, deliverables, troubleshooting procedures, and assessment criteria.
- Oversee 10+ analog-electronics areas, including feedback, filters, oscillators, converters, current mirrors, differential amplifiers, and cascode circuits; guide simulation through measurement and reporting.

Electric Circuits Laboratory Assistant
Sep 2025 - Present

Institut Teknologi Bandung | Part-time | Bandung, Indonesia / On-site

- Mentor 60+ students in circuit theory, LTspice, oscilloscope operation, filters, op-amps, and debugging; validate theoretical, simulated, and measured behavior.
- Selected as the youngest and only third-year assistant on a 17-member team; voted Favorite Assistant.

Research Assistant in Partial-Discharge Detection
Feb 2025 - Feb 2026

Institut Teknologi Bandung | Part-time | Bandung, Indonesia / On-site

- Conducted MATLAB modeling and Verilog signal-processing research under Prof. Nana Sutisna.
- Built a 16-bit peak detector with CSV-to-MEM conversion for FPGA simulation; designed an 8,192-point decimation-in-time FFT for high-resolution frequency analysis.

Digital Systems Assistant Lecturer
Sep 2025 - Jan 2026

Institut Teknologi Bandung | Part-time | Bandung, Indonesia / On-site

- Instructed 50+ students in VHDL, RTL methodology, FSMs, processes, and entity-architecture structure; taught ModelSim testbenches, waveform inspection, and timing/logic debugging.
- Supervised CORDIC and 64-point FFT projects, including synthesis, pin assignment, timing/resource analysis, and Cyclone IV deployment in Quartus Prime Standard 24.1.

Digital Systems Laboratory Assistant
Sep 2025 - Jan 2026

Institut Teknologi Bandung | Part-time | Bandung, Indonesia / Remote

- Led FPGA laboratory sessions for 40+ students using Intel Quartus Prime and Terasic DE10-Lite; guided frequency dividers, FSMs, UART, and live hardware/code debugging.
- Selected as the youngest and only third-year assistant on a 15-member team.

Programming Problem Solving Course Assistant
Feb 2026 - Jun 2026

Institut Teknologi Bandung | Full-time | Bandung, Indonesia / On-site

- Taught 40+ students C programming and Arduino integration through lectures, case studies, practical exercises, and team projects.
- Guided problem decomposition, technical tradeoffs, testing, and debugging; supported assignments and examinations with constructive feedback and responsible programming practices.

SELECTED ENGINEERING PROJECTS

Signed INT8 GAN Accelerator - RTL to GDSII
Implemented a structural 4-by-4 processing-element array with SRAM buffers and a serial host interface on GF180MCU. Verified RTL and post-route behavior bit-for-bit across a 784-pixel image; reported 0.8 GOPS at 25 MHz and zero DRC, LVS, XOR, and antenna violations. Accepted paper at APSIPA ASC 2026, Paper ID 1571326725.

Technologies: SystemVerilog, GF180MCU, SRAM, physical implementation, static timing analysis.

IGOR - Indonesian Gated Onboard Robotics
Designed an eight-lane FPGA neural accelerator for mobile robotics: 138.32 microseconds inference at 50 MHz, 256 reference vectors with zero mismatches, and 98/98 successful Jetson-FPGA UART requests. The IHP ASIC contribution is an I/O-engine proof of concept; the neural core remains on FPGA.

Technologies: Verilog, Cyclone V, Fixed-point inference, Python, UART, IHP 130 nm

FPGA-Based Programmable GAN
Implemented the MLP-based GAN with teammates I Made Medika and Dennis. Created an image GAN for MNIST and a MIDI GAN that generates music from random noise.

Technologies: PYNQ-Z1, SystemVerilog, Python, WGAN, MLP

Three-Qubit Quantum Fourier Transform Tapeout Contribution
Contributed DSP control and digital-track integration with Team D22: Chipster in IEEE SSCS Chipathon 2025. Supported a three-qubit QFT design and completed GF180 tapeout through an agentic workflow spanning reference retrieval, RTL generation, validation, and physical-design feedback.

Technologies: FastAPI, Pydantic, PyTorch, LangChain, ChromaDB, OpenLane 2

TECHNICAL SKILLS

Digital design and verification: SystemVerilog, Verilog, VHDL, RTL microarchitecture, pipelining, FSMs, self-checking testbenches, SVA, functional coverage, ModelSim, Quartus, FPGA validation.

IC implementation: LibreLane, OpenLane, OpenROAD, GF180MCU, static timing analysis, DRC, LVS, antenna closure, Cadence Virtuoso.

Edge AI, embedded and DSP: Python, TensorFlow, YAMNet, PyTorch, Raspberry Pi, ESP32-S3, INMP441, I2S, USB CDC, UART, MATLAB, fixed-point arithmetic, FFT, CORDIC.

Software and automation: C, C++, JavaScript, SQL, FastAPI, Pydantic, LangChain, ChromaDB, Power Apps, Power Automate, Excel, MQTT, InfluxDB, Grafana.

ADDITIONAL EXPERIENCE

Tiny Tapeout | RTL Functional Verification Engineer | Freelance, Remote | Oct 2026. Completed a focused APB-controlled SPI Master verification engagement using SystemVerilog, SVA, functional coverage, and Python regressions; documented results and remaining verification gaps.

Leadership and community: HME ITB web developer (Nov 2024 - Feb 2025); PPSN local coordinator (Oct 2023 - Jan 2024) and teaching officer (Jul - Aug 2024); KMB Dhammanano ITB spiritual division and tutoring (Jul 2024 - Apr 2025). Full organization history: wlmoi.vercel.app/#organizations.

AWARDS & SELECTED CERTIFICATIONS

2nd Place - PERURI Chip Hackathon 2026 (October 2026); 1st Place - PERURI Chip Hackathon Minigames 2026 (October 2026); Most Outstanding Electrical and Electronics Engineering Student 2026 (2026); Ganesha Awardee (2024 & 2025); Winner - IEEE Chipathon 2025 (2025).

Selected certifications: SoC Verification using SystemVerilog; IEEE Self Sensing Processor Systems; Toyota Industrial Instrumentation, Level I; 72-hour Winter School in Advanced Electronics (TUSUR/ITB, 2025); HackerRank JavaScript (Intermediate); EFSET English, 81/100 (Proficient). Full credential library: wlmoi.vercel.app/#certifications.

Languages: Indonesian (native / bilingual); English (full professional); Chinese (elementary).